Is Interpretability a Warrant for Trust in Machine-Learning Intrusion Detection? A Measurement Study

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***Abstract*—A fast-growing line of work equips machine-learning intrusion detection systems (ML-IDS) with explanations, arguing that the resulting interpretability lets an analyst *trust* the detector. We ask, for ML-IDS specifically, what an explanation must satisfy for such trust to be *warranted*. Building on prior work on warranted trust, explainer disagreement, and model multiplicity, we formalize three necessary properties of the explanation map: *fidelity* (the interpretable artifact decides as the deployed model), *determinacy* (the explanation is fixed by the decision, not by the choice of attribution method), and *groundedness* (it reflects the task, not the incidental model instance). On five modern datasets (2015–2023) all three fail: an interpretable surrogate loses up to 0.35 F_1; cross-family explainers agree on as little as 0.15 of the features they show; and equally accurate detectors disagree on nearly half their top features. The failure carries security weight rather than merely epistemic weight: on stable, *correctly detected* true attacks, two equally accurate detectors' explanations drive *opposite* analyst actions, dismiss versus escalate, for up to 0.87 of flows on CICIDS2017 (a simulated-analyst upper bound), so the same true positive is triaged either way depending on which equally-defensible model was deployed. A synthetic positive control confirms the metrics are satisfiable, not rigged to fail. Two honest negatives, a non-transferring attack and a circular, non-transferable, gameable reliability gate, show the warrant is not cheaply recovered. We distill the study into threat-aware evaluation criteria for such claims. Interpretability is not, by default, a warrant for trust in ML-IDS, and recovering that warrant is an open problem.**

I. Introduction

Machine-learning intrusion detection systems (ML-IDS) are opaque, and opacity is a problem in a setting where a human analyst must decide whether to act on an alarm and is accountable for that decision. The dominant response is *explainable* intrusion detection: attach to each alarm an explanation, typically a ranked list of the input features most responsible for it, and argue that the analyst can now understand, verify, and *trust* the detector. Recent work reports that explanations raise analyst trust, cut false-positive dismissals, and speed triage, and surveys frame interpretability as the route to trustworthy and accountable security AI [1].

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This paper asks a question the literature leaves implicit. Trust, in a security setting, is not a feeling; it is a warrant for action under an adversary. If an analyst escalates or suppresses an alarm *because* of the explanation, then the explanation is part of the trusted base of the security operation. So we must ask: **what must be true of an explanation for the trust placed in it to be well-founded?** The explainable-IDS literature measures whether explanations *increase* trust (through user studies, satisfaction scores, triage-time reductions); it rarely asks whether that trust is *warranted*.

We take up the question through systematic measurement. Our contributions are:

- We **formalize interpretability-based trust** for ML-IDS (Section IV): for attribution-driven workflows in which an analyst cannot adjudicate among legitimate surrogate, explainer, and training-instance choices, we derive three audit conditions the explanation must satisfy for that trust to be warrant-invariant, *fidelity*, *determinacy*, and *groundedness*, building on prior notions of warranted trust, explainer disagreement, and model multiplicity rather than claiming these phenomena as new.
- We provide, to our knowledge, the first **joint measurement** of all three, on **five modern NIDS datasets** and **five standard explainers (gradient saliency, input×gradient, integrated gradients, SHAP, LIME)**, under one action-centered framework, and find all three fail with multi-seed statistics (Section V). Explainers from different families agree on as little as 0.15 of the features they present, even at high sampling budgets.
- We show the failure can have **security consequences** (Section VI): on correctly-detected true attacks, where detection is stable, equally-good detectors' explanations induce *opposite* analyst actions at a dataset-dependent rate, up to 0.87 of flows on CICIDS2017 (the most realistic corpus) and near-zero on the easiest task, and there make a true attack read benign 0.29 to 0.35 of the time, so where the effect is strong, mis-founded trust turns whether a real attack is acted on into an accident of the deployed model. These simulated-analyst rates are upper bounds (Section VI).
- We map **how hard the failure is to fix** with two honest negatives: an explanation-manipulation attack does not transfer to tabular IDS (Section VII), and the most

natural per-alarm reliability gate, though it scores well in-distribution, is circular, non-transferable, and adversarially gameable (Section VIII). Trust is not cheaply recovered.
- We distill **threat-aware evaluation criteria** (Section IX) that a defensible interpretability-for-trust claim in security must meet.

We show that the inference "the model is interpretable, therefore its alarms can be trusted" is, as currently supported, unsound; that the failure has a concrete security cost; and that recovering the warrant is an open problem, not a matter of bolting on a confidence score. This is a foundations-of-security question: the conditions under which a security control may be relied upon.

II. Motivation and Setting

Consider a security operations center where an ML-IDS raises an alarm on a network flow and presents the analyst with the five features most responsible, say a large backward packet length and a short flow duration. The analyst, trained to read such explanations, judges the alarm consistent with a known attack and escalates; or, seeing features that look like ordinary traffic, dismisses it as a false positive. Vendors and recent research encourage exactly this workflow, and human-factors studies report that analysts act on explanations and trust systems more when explanations are present.

The workflow embeds an assumption: that the shown features are a reliable account of why the detector fired. If they are not, the analyst's judgment is untethered from the detector's actual behavior, and worse, from the ground truth of whether the flow is an attack. Three ways the account can be untethered motivate our properties. It can reflect a *different model* than the one deployed (if interpretability is delivered by a surrogate). It can be an artifact of the *explanation method* rather than the decision (if a different, equally standard method would have shown different features). And it can be an accident of the *particular trained model* rather than the task (if an equally accurate retraining would explain differently). We make these precise next and then measure them.

Scope. We study network intrusion detection on tabular flow features, the setting where explanations are usually feature attributions over named quantities. We consider standard attribution explainers rather than prototype- or rule-based methods; extending to those is a direction our criteria call for. We treat detection itself as given and study the explanation layer that sits on top of it.

III. Background and Related Work

Attribution methods. The explanations attached to ML-IDS are almost always feature attributions. The dominant methods split into gradient-based, LIME [2], integrated gradients [3], plain saliency [4], SmoothGrad [5], DeepLIFT [6], layer-wise relevance [7], and perturbation- or game-theoretic ones, LIME and SHAP [8], with rule-form variants such as Anchors [9] and, in vision, Grad-CAM [10]. We study five representative members; Molnar [11] surveys the space.

Critiques of ML-IDS evaluation. A tradition of foundational scrutiny warns that ML security results often do not mean what they appear to. Sommer and Paxson [12] argued that the closed-world assumptions of ML evaluation break in operational intrusion detection; Arp et al. [13] catalogue pervasive experimental pitfalls (spatial and temporal snooping, base-rate neglect, inappropriate baselines); Axelsson's base-rate analysis [14] shows why laboratory detection rates mislead at realistic prevalence; and broad surveys map the role and fragility of ML in security [15], [16]. These critiques target *accuracy* claims and their datasets. Ours targets a newer claim, *interpretability for trust*, that the accuracy critiques do not address.

Explainable intrusion detection. A large recent literature applies attribution methods to ML-IDS and argues they enhance analyst trust and accountability [1], including work that evaluates explanation methods specifically for security [17] and systematizes explainable ML across security applications [18], [19]. Closest to our empirical setting, E-XAI [20] benchmarks SHAP and LIME on NIDS datasets with stability, robustness, and completeness metrics; a survey of explainable IDS [21] calls for stakeholder-specific evaluation, and a field deployment of xAI tooling in a security operations center [22] reports the same need from the analyst side; and constructive proposals design action-oriented explanations, e.g. feasible-action contracts for intrusion triage [23]. These works measure explanation *quality* or user acceptance, or build explanations. We are orthogonal and adversarial: taking their metrics as given, we ask whether the trust they induce is *warranted*, jointly test the surrogate, method, and model choices an analyst cannot adjudicate, and trace the action-level consequence, none of which the above do together.

Fragility, disagreement, and multiplicity. In vision and NLP, explanations are known to be fragile and unreliable: Ghorbani et al. [24] perturb inputs to change saliency while fixing the prediction; Dombrowski et al. [25] manipulate explanations arbitrarily; Slack et al. [26] fool LIME and SHAP with adversarial scaffolding, a technique Senevirathna et al. [27] extend to deceive NIDS operators (our honest-negative attack in Sec. VII is a different, bounded input-perturbation formulation and does *not* transfer, which we report as a boundary result, not as robustness); sanity checks show some saliency maps are independent of the model [28]; saliency methods are provably unreliable under input shifts [29], [30]; and attribution methods disagree systematically [31], [32]. Our groundedness property is the security-relevant face of *model multiplicity*: many equally-accurate models exist (the Rashomon effect [33], [34]) and they can differ arbitrarily in their per-example predictions and explanations [35], [36]. These results are about general ML; we ask what they mean for the *security* claim that interpretability licenses trust, connect them to a formal trust condition, and show they carry a concrete security consequence.

Interpretability foundations. Rudin [37] distinguishes intrinsically interpretable models from post-hoc explanations and

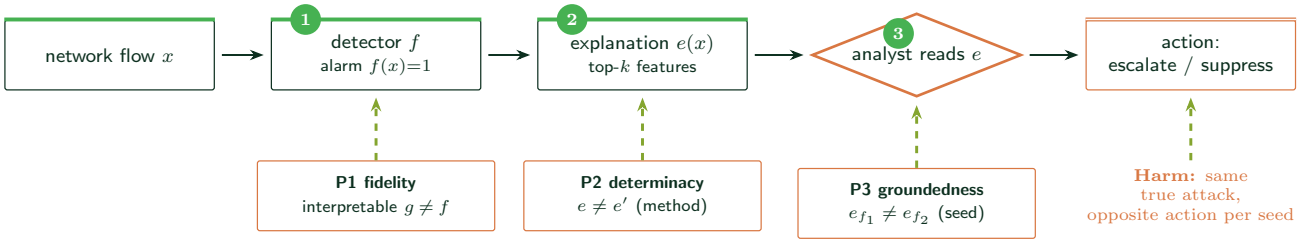


Fig. 1. The analyst-trust pipeline in explainable ML-IDS, and the three points where interpretability-based trust breaks. The detector f raises an alarm; an explanation $e(x)$ is computed; the analyst reads it and acts. Trust fails if the interpretable artifact does not decide as f (P1), if the shown reason changes with the attribution method (P2), or if it changes with the training seed (P3). The security harm: on the same correctly-detected true attack, equally-good detectors' explanations drive opposite analyst actions.

warns against the latter for high-stakes decisions; the broader debate concerns what interpretability even means [38], [39], how explanation functions socially [40], and how trust in AI should be formalized [41] and engineered for [42]. We build directly on Jacovi et al.'s notion of warranted trust; our contribution is not to introduce it but to specialize it to explanation-conditioned IDS action and give operational invariance checks. Our fidelity property makes Rudin's distinction operational for security: a post-hoc or distilled "interpretable" artifact that does not decide as the deployed model does cannot ground trust in that model. Our detectors include the Kolmogorov-Arnold network [43], marketed as interpretable-by-design, precisely to test that claim.

Adversarial ML in security. That ML models are brittle to adversarial input is foundational [44], [45], [46], and realistic evasion of NIDS is an active concern [47]. We consider a distinct target, the *explanation* an analyst reads rather than the detector's label, and find (unlike the detector case) that manipulating it on tabular IDS is not straightforward.

IV. What Interpretability-Based Trust Requires

We model an ML-IDS as a decision function $f : \mathcal{X} \to \{0,1\}$ over network observations $x \in \mathcal{X}$, where $f(x) = 1$ raises an alarm. An *explanation method* is a map $e : \mathcal{X} \to \mathcal{E}$ producing, for each x, an artifact $e(x)$, in practice a ranked set $\mathrm{top}_k(e, x)$ of the k features held responsible, presented to the analyst. An analyst policy a decides, given the alarm and the explanation, whether to act.

Definition 1 (Interpretability-based trust)**.** *An analyst* trusts f through e *when the analyst's action policy depends on $e(x)$: the analyst escalates or suppresses the alarm $f(x)$ on the basis of $e(x)$, not on f alone. The trust is* well-founded *if $e(x)$ is a reliable signal of the grounds of the decision $f(x)$, i.e. acting on e does not systematically differ from acting on the true reasons for $f(x)$.*

Because $e(x)$ enters the analyst's decision, it is part of the trusted base. Well-foundedness therefore requires e to carry genuine information about f's decision and about the task. We measure the overlap of two explanations by the Jaccard agreement of their top-k feature sets,

$$J_k(e, e'; x) = \frac{|\mathrm{top}_k(e, x) \cap \mathrm{top}_k(e', x)|}{|\mathrm{top}_k(e, x) \cup \mathrm{top}_k(e', x)|} \in [0, 1], \qquad (1)$$

and we isolate three necessary properties, each parameterized by a tolerance ε.

Property 1 (Fidelity)**.** *The interpretable artifact the analyst reasons about must decide as the deployed model does. If interpretation is delivered through a surrogate g (distilled or intrinsic), fidelity requires*

$$\Pr_x\big[g(x) \neq f(x)\big] \;\leq\; \varepsilon_{\mathrm{fid}}. \qquad (2)$$

A surrogate that violates (2) *explains a different classifier than the one raising the alarm.*

Property 2 (Determinacy)**.** *The explanation must be fixed by the decision, not by the choice of attribution method. For a family $\mathcal{M}$ of equally-justified explainers, determinacy requires*

$$\min_{e, e' \in \mathcal{M}} \; \mathbb{E}_x\big[J_k(e, e'; x)\big] \;\geq\; 1 - \varepsilon_{\mathrm{det}}. \qquad (3)$$

If (3) *fails, the reason shown to the analyst is a property of the tool, not the decision.*

Property 3 (Groundedness)**.** *The explanation must reflect the task, not the incidental model instance. For models f_1, f_2 drawn from the same training procedure (differing only in seed) with $|F_1(f_1) - F_1(f_2)| \leq \delta$, groundedness requires*

$$\mathbb{E}_x\big[J_k(e_{f_1}, e_{f_2}; x)\big] \;\geq\; 1 - \varepsilon_{\mathrm{grnd}}. \qquad (4)$$

If (4) *fails while δ is negligible, the explanation reflects an accident of training, not the task.*

Table I collects the notation.

We make the necessity precise through an *invariance* view of warrant, which sidesteps the (unavailable) ground truth of the "true reason" for a decision. An analyst who trusts f through e faces choices they cannot principled-ly adjudicate: which interpretable surrogate to read, which attribution method to run, and which of many equally-accurate trained models was deployed. Write $\mathcal{C}$ for this set of legitimate choices and e^c for the explanation under choice $c \in \mathcal{C}$.

TABLE I
NOTATION.

Symbol	Meaning
$f : \mathcal{X} \to \{0, 1\}$	deployed IDS decision (alarm $= 1$)
$e(x),\ \mathrm{top}_k(e, x)$	explanation and its top-k features
g	interpretable surrogate presented to the analyst
J_k	top-k Jaccard agreement, Eq. (1)
$\mathcal{M}$	family of attribution methods (5 in our study)
$S(x)$	explanation reliability score, Eq. (5)
$\varepsilon_\bullet$	per-property trust tolerance

TABLE II
WHICH EXPLANATION FAMILIES CAN, IN PRINCIPLE, SATISFY EACH TRUST PROPERTY. ● STRUCTURALLY PROVIDES IT, ◐ CONDITIONALLY, ○ AT RISK AND MUST BE MEASURED. NO FAMILY SATISFIES ALL FOUR FOR FREE.

Explanation family	fidelity	determ.	ground.	faithf.
gradient attribution	●	○	○	○
SHAP / LIME (perturb.)	●	○	○	◐
global surrogate	◐	●	◐	◐
intrinsic / KAN	●	●	○	○
rule / concept	●	◐	◐	◐

Definition 2 (Warrant invariance)**.** *Trust in f through e is* warrant-invariant *if the analyst's action is stable across legitimate choices: for an action policy a depending on e through top_k, $\Pr_x\left[\exists\, c, c' \in \mathcal{C} :\ a(e^c(x)) \neq a(e^{c'}(x))\right] \leq \eta$ for a small η.*

Proposition 1. *If any of fidelity* (2)*, determinacy* (3)*, or groundedness* (4) *is violated by more than its tolerance ε, then trust is not warrant-invariant: there is a positive-measure set of flows on which two legitimate choices induce different top-k explanations, hence different actions for any a that separates those top-k sets.*

Proof sketch. Each property bounds the disagreement between two legitimate choices. If determinacy fails, $\mathbb{E}_x[J_k(e, e'; x)] < 1 - \varepsilon_{\mathrm{det}}$ for two methods $e, e' \in \mathcal{M}$; since $J_k \leq 1$, a positive-measure set has $\mathrm{top}_k(e, x) \neq \mathrm{top}_k(e', x)$, and any a that maps these distinct sets to distinct actions violates Definition 2 with $\eta < \varepsilon_{\mathrm{det}}$. The groundedness case is identical with c, c' two equal-accuracy seeds. For fidelity, a surrogate g with $\Pr[g \neq f] > \varepsilon_{\mathrm{fid}}$ gives, on a positive-measure set, an explanation of a decision the deployed model did not make, so the choice "read g" versus "read f" changes the grounds the analyst acts on. Thus each violation exhibits an unadjudicated choice that flips the action. □

Proposition 1 is deliberately about invariance, not correctness: two methods can agree and both mislead, so agreement is not sufficient for a *correct* explanation. But disagreement is sufficient for an *unwarranted* one, because the analyst has no principled basis to prefer either choice. The necessity is scoped to the policy class, not universal: the proposition shows that *some* action policy sensitive to top_k diverges under unadjudicated choices, and Section VI exhibits two *learned* such policies that do; it does not assert that every conceivable analyst policy would. A policy that ignored the explanation would be trivially invariant, but then interpretability plays no role in the trust it is invoked to support. We now show current practice violates all three properties, then that the violation carries a security cost and resists a cheap fix.

A. Which explanation approaches could warrant trust?

Before measuring, we systematize the landscape: which *families* of explanation could, in principle, satisfy each property (Table II)? Post-hoc attributions (gradient [4], [3], perturbation-based SHAP and LIME [8], [2]) are computed *from* a fixed deployed model, so they cannot violate fidelity, but they are method-dependent (determinacy at risk) and model-instance-dependent (groundedness at risk), and faithfulness is not guaranteed. Global surrogates (trees, sparse linear models) can be faithful only up to their approximation of the deployed model, trading fidelity against readability (Section V). Intrinsically interpretable models [37] (and interpretable-by-design networks such as the KAN [43]) escape the surrogate gap by construction, but only if the deployed model *is* the interpretable one, which sacrifices the capacity practitioners deploy for, and, as we show, a KAN's readable symbolic form is not faithful to the trained KAN. Rule- and concept-based methods [9], [48] are more stable but coarser. No family is free of all four risks; the table makes explicit where each must be checked.

V. THE THREE PROPERTIES FAIL

Setup. We evaluate on *five* modern NIDS datasets (Table III), each with named, interpretable flow or connection features so that a feature attribution is a claim about a real network quantity rather than an opaque coordinate. We deliberately span heterogeneous capture settings, attack taxonomies, and class balances so that our negative findings cannot be blamed on a single corpus. Each dataset is cast as binary normal-versus-attack detection with standardized features and stratified $70/30$ train/test splits. Our detectors are a multilayer perceptron (MLP) and a Kolmogorov-Arnold network (KAN, whose learnable univariate edges are explicitly marketed as interpretable); explanations are gradient saliency, input×gradient, integrated gradients, KernelSHAP, and LIME. We further confirm determinacy, groundedness, and faithfulness on the tree ensembles that dominate tabular NIDS, random forests and gradient boosting explained by exact TreeSHAP (Table VII). Results are over three seeds (properties P1, P2) and four seeds (P3); we report mean and standard deviation. Table IV and Fig. 2 summarize the findings; each property should score 1.0 for trust to hold, and none does on any dataset. Appendix B maps every reported quantity to the script and result file that produce it.

Dataset descriptions. We chose the five corpora to span 2015–2023 of NIDS practice and five distinct deployment regimes, so that any property that held only under one capture pipeline

TABLE III

THE FIVE MODERN NIDS DATASETS. ALL HAVE NAMED, INTERPRETABLE FLOW/CONNECTION FEATURES, SO AN ATTRIBUTION IS A CLAIM ABOUT A REAL NETWORK QUANTITY. "ATTACK RATE" IS THE POSITIVE-CLASS FRACTION AFTER OUR 40K STRATIFIED SUBSAMPLE.

Dataset	Year	Features	Attack rate	Capture setting	Attack families
UNSW-NB15 [49]	2015	34	0.68	IXIA PerfectStorm testbed	9 families: DoS, exploits, reconnaissance, fuzzers, shellcode, worms, backdoors, generic, analysis
CICIDS2017 [50], [51]	2017	52	0.17	campus network flows	DoS/DDoS, web attacks, infiltration, PortScan
5G-NIDD [52]	2022	48	0.61	live 5G core testbed	DoS, port and vulnerability scanning
TON-IoT [53]	2020	38	0.76	IoT/IIoT edge testbed	scanning, DoS/DDoS, backdoor, ransomware, injection, XSS, password, MITM
CIC-IoT-2023 [54]	2023	39	0.97	IoT testbed, 105 devices	DDoS, DoS, Mirai, reconnaissance, spoofing, web, brute force

would be exposed. *UNSW-NB15* [49] is generated by the IXIA PerfectStorm tool at ACCS (2015) and mixes normal traffic with nine synthetic attack families (fuzzers, analysis, backdoors, DoS, exploits, generic, reconnaissance, shellcode, worms); we use the 34 numeric/protocol flow features. It is the most balanced corpus (attack rate 0.68) and remains a de facto ML-IDS benchmark. *CICIDS2017* [50] captures five days of a realistic campus network with 80-plus CICFlowMeter statistics; we use the cleaned release of Engelen et al. [51], which repairs the labeling and feature-extraction errors of the original, and keep 52 features. It is the most imbalanced (0.17) and its labeled attack types (DoS/DDoS, web attacks, infiltration, PortScan) let us analyze harm per attack type (Sec. VI). *5G-NIDD* [52] is captured on a live 5G test network (2022) and is the closest to the next-generation settings where ML-IDS is now marketed; its 48 features describe DoS and scanning attacks against 5G services. *TON-IoT* [53] is a telemetry corpus from a medium-scale IoT/IIoT edge testbed (2020); after dropping identifier columns (IP addresses, ports, timestamps) that leak host identity rather than behavior, we keep 38 connection features. Its rich attack taxonomy (scanning, DoS/DDoS, backdoor, ransomware, data injection, XSS, password, MITM) makes it, with CICIDS2017, the second corpus on which we measure per-attack-type harm. *CIC-IoT-2023* [54] is the most recent corpus, captured on a large IoT testbed of 105 real devices (UNB, 2023); we use its 39 named packet and flow features. It is heavily attack-dominated (rate 0.97, DDoS-heavy), so detection is easy and the detector saturates; we add it specifically to test whether the failures persist on current, next-generation data, and the unconfounded ones do (determinacy 0.51, groundedness 0.49; its near-zero faithfulness we read only as consistent, being confounded by saturation, Sec. V).

Fidelity fails (Property 1). We take a KAN-IDS, whose per-edge functions are its interpretability selling point, and extract the interpretable form its proponents invoke, a low-order symbolic surrogate of each edge. Detection F_1 drops by 0.08 ± 0.01 on UNSW-NB15, 0.35 ± 0.03 on CICIDS2017, 0.11 ± 0.01 on 5G-NIDD, and 0.34 ± 0.04 on TON-IoT. On

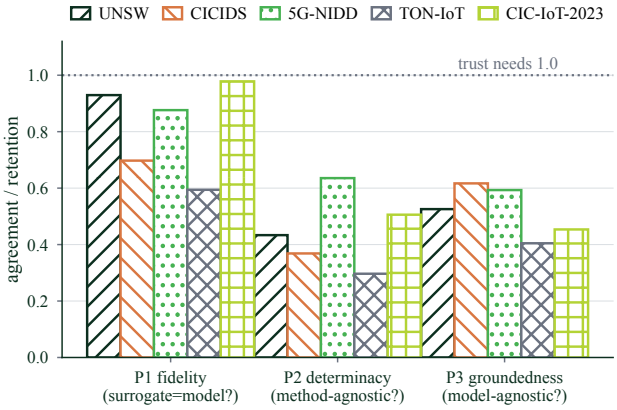


Fig. 2. Each necessary property, measured on five modern NIDS datasets. A sound trust signal requires every bar at 1.0; all fall short on every dataset.

CICIDS2017 the interpretable form collapses from $F_1 = 0.92$ to 0.62 (Fig. 3): the readable model is nearly a different, and far worse, detector. A post-hoc check agrees: keeping only the top-5 attributed features reproduces the 5G-NIDD detector's decision on just 0.63 of flows, so the reason does not even suffice for the decision. The analyst reasons about a model that is not the one guarding the network.

This is not special to the KAN. Fitting standard *global* interpretable surrogates to mimic a deployed MLP-IDS reveals a fidelity-readability frontier (Table V): a higher-capacity depth-4 tree matches the deployed decisions well (0.98 agreement on CICIDS2017) but is barely more readable than the model, whereas the genuinely readable renderings, a depth-2 tree or a sparse logistic model, lose detection quality (F1 falls from 0.93 to 0.72 and 0.84 on CICIDS2017). Fidelity is thus attainable only by surrendering the readability the trust claim depends on.

Determinacy fails (Property 2). For the same detector and the same flagged-malicious flows, we measure how much the top-5 features agree across explainers. Among the three gradient-family methods, mean agreement is only 0.45 ± 0.017 (UNSW-NB15), 0.36 ± 0.005 (CICIDS2017), 0.53 ± 0.079 (5G-

TABLE IV

EACH NECESSARY PROPERTY OF INTERPRETABILITY-BASED TRUST, ON FIVE MODERN NIDS DATASETS (2015–2023), OVER THREE SEEDS (P1, P2) AND FOUR SEEDS (P3), MEAN±STD. EVERY QUANTITY IS AN AGREEMENT OR RETENTION THAT MUST BE ≈ 1.0 FOR TRUST TO BE WELL-FOUNDED; DETERMINACY AND GROUNDEDNESS FALL SHORT ON EVERY DATASET.

Property	Quantity	UNSW	CICIDS	5G-NIDD	TON-IoT	CIC-IoT23
P1 Fidelity	deployed vs. interpretable-surrogate F_1 drop ↓	0.08±0.01	**0.35±0.03**	0.11±0.01	**0.34±0.04**	0.01±0.01
	top-5 attribution sufficiency (agree w/ decision)	0.92	0.89	0.63	0.87	0.96
P2 Determinacy	mean top-5 agreement, 3 gradient methods	0.45±0.02	0.36±0.01	0.53±0.08	0.37±0.06	0.51±0.01
	mean top-5 agreement, all 5 explainers	0.35	0.34	0.50	0.32	0.36
P3 Groundedness	detection F_1 across 4 seeds	.95±.001	.93±.002	1.0±.000	.99±.000	.99±.000
	cross-seed top-5 explanation agreement	0.53±0.01	0.62±0.03	0.56±0.03	**0.37±0.03**	0.49±0.03

TABLE V

THE FIDELITY-READABILITY FRONTIER: EACH CELL IS DECISION AGREEMENT WITH THE DEPLOYED MLP-IDS (HIGHER IS MORE FAITHFUL) / DETECTION F_1, FOR GLOBAL INTERPRETABLE SURROGATES. READABLE SURROGATES (DEPTH-2 TREE, SPARSE LOGIT) ARE LESS FAITHFUL OR DETECT WORSE.

Surrogate	UNSW	CICIDS	5G	TON	CIC23
tree (d=4)	.96/.94	.98/.94	1.0/1.0	1.0/.99	.99/.99
tree (d=2)	.94/.92	.94/.**72**	.98/.98	.94/.96	.98/.99
sparse logit	.96/.94	.96/.**84**	.99/.99	.96/.97	.99/.99



Fig. 3. Fidelity fails. The interpretable surrogate of a KAN-IDS detects markedly worse than the deployed model, collapsing on CICIDS2017. The analyst reasons about the lower bar.

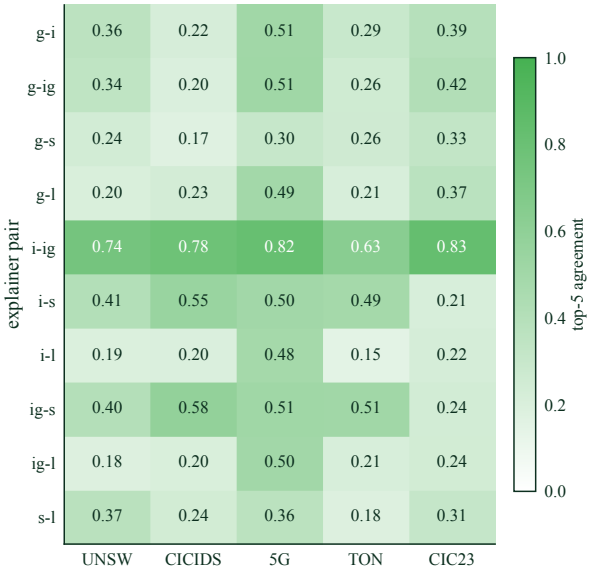


Fig. 4. Determinacy across five explainers (high sampling budgets): top-5 agreement for all ten pairs on each dataset. Only the two input×gradient-based methods (i-ig) agree; every cross-family pair is light (low). The reason shown to the analyst is largely a tool artifact. The underlying numbers for all ten pairs are in Appendix A.

NIDD), and 0.37±0.06 (TON-IoT). This is not an artifact of using only gradient methods, nor of under-sampling: adding the two most deployed model-agnostic explainers, SHAP and LIME, at high sampling budgets, the mean five-explainer agreement stays at 0.32–0.50 (Fig. 4), and *cross-family* agreement is lower still: gradient saliency and SHAP agree on only 0.17–0.31 of the top-5 features. The reason presented to the analyst is largely an artifact of which equally-defensible tool the vendor shipped.

Groundedness fails (Property 3). We train four detectors that differ only in random seed. Their detection performance is statistically identical (on UNSW-NB15, $F_1 = 0.95\pm0.001$), yet their top-5 explanations for the same inputs agree on only 0.53±0.005 (UNSW-NB15), 0.62±0.029 (CICIDS2017), 0.56±0.027 (5G-NIDD), and 0.37±0.03 (TON-IoT) of features. An analyst handed one detector's explanation is trusting an accident of training that an equally good detector would contradict.

All three necessary properties fail, with tight seed-to-seed variance, across five modern datasets. The failures are not artifacts of our choices: determinacy agreement stays low across top-k sizes (k=3, 5, 10; e.g. CICIDS2017 0.35–0.47), and groundedness agreement is flat as the ensemble grows from four to ten models (CICIDS2017 $\approx$ 0.46, 5G-NIDD $\approx$ 0.63), so neither failure is rescued by showing more features or averaging more models (Fig. 5). Nor is the disagreement an artifact of the redundancy of flow features: collapsing features into Spearman-correlation clusters, so that an explainer picking a near-duplicate of another's feature is scored as agreeing, raises agreement by at most 0.14 at a near-duplicate threshold ($|\rho| \geq 0.9$) and never above 0.76 (Table VIII); even at an aggressive $|\rho| \geq 0.8$ the peak is 0.82, on the most redundant

TABLE VI

POSITIVE CONTROL: THE METRICS ARE SATISFIABLE, NOT RIGGED TO FAIL. ON A SYNTHETIC TASK WITH 32 DECORRELATED FEATURES AND A LINEARLY SEPARABLE LABEL DRIVEN BY ONLY k OF THEM, ALL THREE PROPERTIES REACH ≈ 1.0 WHEN THE GROUND-TRUTH IS SPARSE (k=5), AND THE MODEL'S TOP-5 SALIENCY RECOVERS THE TRUE INFORMATIVE FEATURES EXACTLY (RECOVERY 1.00). SPREADING THE SIGNAL OVER MORE FEATURES (LARGER k), WHICH REMOVES THE SPARSE, DECORRELATED STRUCTURE THAT INTERPRETABILITY IMPLICITLY ASSUMES, DRIVES THE SCORES DOWN TO THE REAL-NIDS REGIME.

Synthetic regime	P1 suff.	P2 determ.	P3 ground.
sparse (k=5, ceiling)	**0.99**	**0.92**	**1.00**
mid (k=12)	0.76	0.56	0.76
diffuse (k=28)	0.67	0.43	0.44

TABLE VII

THE DETERMINACY, GROUNDEDNESS, AND FAITHFULNESS FAILURES ARE NOT SPECIFIC TO NEURAL DETECTORS: THEY HOLD, IN THE SAME RANGE, ON THE TREE ENSEMBLES THAT DOMINATE TABULAR NIDS, EXPLAINED BY *exact, deterministic* TREESHAP. RANGES ARE ACROSS THE FIVE DATASETS. FOR TREES, DETERMINACY IS MEASURED ACROSS EQUALLY-DEFENSIBLE ATTRIBUTIONS (TREESHAP, LIME, IMPURITY, PERMUTATION), SINCE TREESHAP ALONE IS DETERMINISTIC; A READABLE DEPTH-2 SURROGATE ALSO LOSES FIDELITY (E.G. CICIDS2017 F_1 0.99→0.72), AS FOR THE NEURAL DETECTORS.

Detector family	P2 determ.	P3 ground.	Faithful. (comp.)
Neural (MLP, KAN)	0.36–0.53	0.37–0.62	0.00–0.51
Random forest	0.35–0.48	0.52–0.67	0.00–0.59
Gradient boosting	0.21–0.45	0.44–0.76	0.00–1.00

dataset, still far from the $\approx$ 1.0 a trust signal needs. The under-determination is thus between genuinely distinct feature clusters, not redundant column pairs. Finally, the metrics are not unsatisfiable by construction: on a synthetic positive control with decorrelated features and a *sparse known* ground-truth, all three properties score 0.92–1.00 and the model's top-5 recover the true informative features exactly, and they fall to the real-NIDS regime ($P2{\approx}0.43$, $P3{\approx}0.44$) only as the sparse, decorrelated structure is removed (Table VI). The metrics are therefore satisfiable *and* calibrated; the failures on real NIDS are a property of real explanations, not of the measurement. This also sets the bar empirically rather than as a strawman: a genuinely grounded task reaches 0.92–1.00 here, so the real-NIDS scores of 0.4–0.6 fall well short of the *achievable* ceiling, not merely of an idealized 1.0. The failures are also not specific to neural detectors: on the tree ensembles that dominate tabular NIDS (random forests and gradient boosting), explained by *exact, deterministic* TreeSHAP, determinacy (across TreeSHAP, LIME, impurity, and permutation attributions), groundedness (across seeds), and faithfulness all fail in the same range as for the MLP and KAN (Table VII). The trust that explainable ML-IDS invites is therefore, as currently produced, not well-founded.

A fourth, deeper property: faithfulness. The three properties above concern the *consistency* of an explanation (across surrogate, method, and seed) and are each necessary for warrant. Faithfulness is stronger: even a perfectly consistent

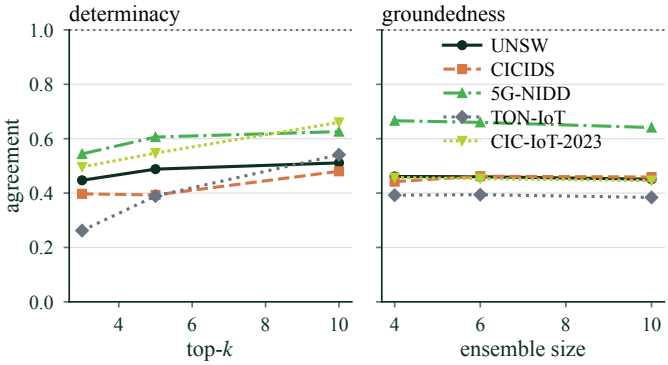


Fig. 5. Robustness of the determinacy and groundedness failures. Agreement stays low as the shown top-k grows (left, k=3, 5, 10) and as the ensemble grows (right, E=4, 6, 10), far below the 1.0 a trust signal needs. The failures are not artifacts of k or ensemble size.

TABLE VIII

THE DETERMINACY (P2) AND GROUNDEDNESS (P3) FAILURES ARE NOT A FEATURE-REDUNDANCY ARTIFACT. "RAW" IS PER-FEATURE TOP-5 JACCARD; THE "$|\rho| \geq \tau$" COLUMNS FIRST COLLAPSE FEATURES INTO SPEARMAN-CORRELATION CLUSTERS, SO AN EXPLAINER THAT PICKS A NEAR-DUPLICATE OF ANOTHER'S FEATURE COUNTS AS AGREEING. EVEN AT AN AGGRESSIVE τ=0.8 NO VALUE REACHES THE $\approx$ 1.0 A TRUST SIGNAL NEEDS. RAW VALUES ARE SINGLE-SEED AND MATCH TABLE IV WITHIN SEED VARIANCE.

	Grouping	UNSW	CICIDS	5G	TON	CIC23
P2	raw	0.49	0.39	0.61	0.39	0.55
	$\lvert\rho\rvert \geq 0.9$	0.60	0.46	0.71	0.43	0.57
	$\lvert\rho\rvert \geq 0.8$	0.69	0.51	0.79	0.63	0.58
P3	raw	0.56	0.61	0.62	0.39	0.48
	$\lvert\rho\rvert \geq 0.9$	0.68	0.63	0.76	0.51	0.49
	$\lvert\rho\rvert \geq 0.8$	0.75	0.69	**0.82**	0.57	0.50

explanation could be wrong about the model. We treat it separately because, unlike the agreement properties, it compares the explanation to the model's actual behavior rather than to another explanation. One might hope the shown features are the ones the model *uses*. They are not. We measure the two standard faithfulness metrics [32], comprehensiveness (the drop in attack-probability when the top-5 features are removed) and sufficiency (how much the top-5 alone retain), on all five datasets. Removing the top-5 attributed features barely moves the decision: comprehensiveness is only 0.18 (UNSW-NB15), 0.16 (5G-NIDD), 0.04 (TON-IoT), and -0.001 on CIC-IoT-2023, and reaches 0.51 only on CICIDS2017. So on four of five datasets the features the analyst is shown (via gradient saliency) are largely not the ones driving the alarm (Table X). The near-zero value on CIC-IoT-2023 is a boundary case we flag rather than lean on: its detector is saturated (attack rate 0.97, attack-probability $\approx$ 1.0 on flagged flows), so no small feature set is comprehensive, and faithfulness here is confounded with task triviality; the unconfounded evidence on this corpus is its determinacy (0.51) and groundedness (0.49) failures, which do not depend on detector confidence. Extending to all five explainers (Table IX) shows faithfulness

TABLE IX

COMPREHENSIVENESS OF THE TOP-5 EXPLANATION FOR ALL FIVE EXPLAINERS (HIGHER IS MORE FAITHFUL). FAITHFULNESS IS EXPLAINER- AND DATASET-DEPENDENT: NO METHOD IS RELIABLY FAITHFUL ACROSS DATASETS, AND THE MOST-DEPLOYED (SHAP, LIME) SWING WIDELY. †CIC-IOT-2023'S DETECTOR IS SATURATED (ATTACK RATE 0.97), SO NO SMALL FEATURE SET IS COMPREHENSIVE; WE DO NOT READ ITS ZEROS AS AN INDEPENDENT FAITHFULNESS RESULT.

	grad	i×g	int-g	SHAP	LIME
UNSW-NB15	0.17	0.32	0.33	0.35	0.12
CICIDS2017	0.47	0.81	0.74	**0.95**	0.68
5G-NIDD	0.15	0.28	0.30	0.40	0.35
TON-IoT	**0.04**	0.09	0.07	0.34	0.19
CIC-IoT-2023†	0.00	0.00	0.00	0.00	0.00

TABLE X

FAITHFULNESS OF THE TOP-5 EXPLANATION ON FLAGGED ATTACKS, ACROSS FIVE MODERN DATASETS. COMPREHENSIVENESS (DROP IN ATTACK-PROBABILITY WHEN THE TOP-5 ARE REMOVED; HIGHER IS MORE FAITHFUL) IS LOW ON FOUR OF FIVE: THE SHOWN FEATURES ARE LARGELY NOT THE ONES THE MODEL USES. CIC-IOT-2023'S ZERO IS CONFOUNDED BY A SATURATED DETECTOR (SEE TEXT) AND NOT READ AS INDEPENDENT EVIDENCE.

	UNSW	CICIDS	5G	TON	CIC-IoT23
comprehensiveness ↑	0.18	0.51	0.16	0.04	**0.00**
sufficiency gap ↓	0.02	0.37	0.75	0.04	0.00

is real but *unreliable*: it depends on both explainer and dataset. SHAP is the most faithful method but still dataset-dependent, highly faithful on CICIDS2017 (comprehensiveness 0.95) yet only 0.40 on 5G-NIDD and 0.34 on TON-IoT; gradient saliency, the most-deployed explainer, is the least faithful, as low as 0.04 on TON-IoT. No single explainer is faithful across datasets, so an analyst cannot assume the shown features are the operative ones without measuring it per deployment, which no audited paper does (Section IX-A).

VI. THE FAILURE HAS SECURITY CONSEQUENCES

A skeptic may object that under-determination is generic machine-learning hygiene: rename the datasets to medical or credit and the numbers survive. We answer with a consequence that is specifically about *security decisions* (the harm annotated in Fig. 1). Consider flows that are true attacks and that the IDS *correctly* flags; here the detector is right and, across a seed-ensemble, stable. We model an analyst who reads an explanation and decides to escalate (treat as a real attack) or suppress (dismiss as a false positive) with an auditor classifier trained to read attribution vectors; we use two auditor families (logistic regression and a random forest), so the result is not a linearity artifact. Algorithm 1 states the measurement.

The result is stark on CICIDS2017, the most operationally realistic of the five (a 17% attack rate over many attack types). Detection is stable: the seed-ensemble agrees on 0.88 of these flows that they are attacks. Yet the explanation-driven analyst action is not: on 0.48 to 0.87 of the true attacks, two equally-good detectors' explanations induce *opposite* analyst actions,

Algorithm 1: Explanation-induced action divergence on true attacks

Input: seed-ensemble $\{f_1, \ldots, f_E\}$; auditor g mapping an attribution to escalate/suppress; test set with labels

Output: action-divergence rate on correctly-detected true attacks

$T \leftarrow \{x : y(x)=1 \wedge f_i(x)=1 \ \forall i\}$; `// true attacks all models flag`

foreach $x \in T$ **do**

  foreach i **do** $a_i(x) \leftarrow g\big(e_{f_i}(x)\big)$;

  $\mathrm{div}(x) \leftarrow \big[\neg\,(a_1(x)=\cdots=a_E(x))\big]$; `// models split`

return $\frac{1}{|T|}\sum_{x\in T}\mathrm{div}(x)$

and a true attack's explanation reads benign (so the analyst would dismiss it) 0.29 to 0.35 of the time. In other words, the detector agrees a flow is an attack, but whether the analyst *acts* on it is decided by which arbitrary training seed was deployed. This is the security harm: mis-founded trust in the explanation converts a grounded detection into an ungrounded response. The effect is smaller on UNSW-NB15 (divergence ≈ 0.12) and smallest on 5G-NIDD under the logistic auditor (≈ 0.02, where detection is near-perfect and the task easy), and substantial on TON-IoT (0.37 to 0.81 across the two auditors, over its diverse IoT attack taxonomy), so the harm scales with task difficulty and attack diversity, being largest exactly where operational stakes are highest (Fig. 7, Table XI). No adversary is required: the harm follows from the benign under-determination of Section V. The effect is robust to the analyst model: a random-forest auditor amplifies the divergence and keeps it concentrated on the harder datasets (0.87 on CICIDS, 0.81 on TON-IoT), so it is not an artifact of a linear analyst.

The harm is not a coincidence of our auditor; it is a consequence of the groundedness failure, which Proposition 2 states and the measurements below instantiate.

Proposition 2 (Harm from ungroundedness)**.** *Let the analyst action $a(e(x)) \in \{\mathit{escalate}, \mathit{suppress}\}$ depend on the explanation only through top_k, and let π be the probability that a changes when top_k changes on a flow. For two equally-accurate models f_1, f_2 with cross-seed agreement $\mathbb{E}_x[J_k(e_{f_1}, e_{f_2}; x)] = 1 - \varepsilon_{\mathrm{grnd}}$, the action-divergence rate on their shared true positives is at least $\pi\,\varepsilon'_{\mathrm{grnd}}$, where $\varepsilon'_{\mathrm{grnd}}$ is the fraction of flows with disjoint-enough top-k sets.*

Proof sketch. Groundedness failure means a positive-measure set D of flows has $\mathrm{top}_k(e_{f_1}, x) \neq \mathrm{top}_k(e_{f_2}, x)$. On any $x \in D$ where the differing features flip the analyst's read, the two models induce different actions; the measure of such flows is at least π times $|D|$. Since detection is stable on these true positives (both models flag attack), the divergence is entirely explanation-driven. Larger $\varepsilon_{\mathrm{grnd}}$ (worse groundedness) forces larger $|D|$, hence more divergence, which is the monotone

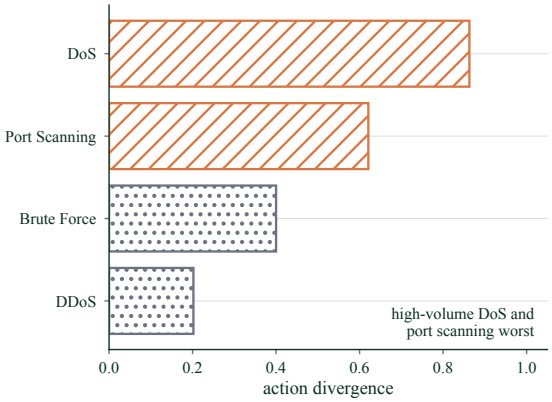


Fig. 6. Action divergence by attack type on correctly-detected true attacks (CICIDS2017). The explanation-induced blind spot is largest on the high-volume DoS and port-scanning flows and smaller on DDoS: mis-founded trust does not spread its risk evenly across attack types.

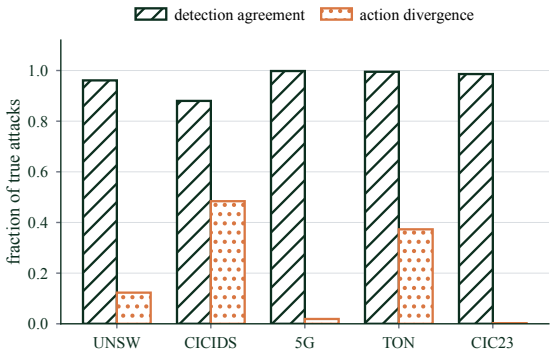


Fig. 7. The security consequences. On correctly-detected true attacks, detection is stable (the ensemble agrees the flow is an attack), yet the explanation-driven analyst action diverges across equally-good models, most on the realistic CICIDS2017. Whether a real attack is acted on becomes an accident of the deployed seed.

trend we observe across datasets (CICIDS, with worse groundedness on hard attacks, diverges most). □

In operational cost. The divergence matters because the two actions have very different costs. Write C_{miss} for the cost of suppressing a true attack and C_{fp} for investigating a false positive, with $C_{\text{miss}} \gg C_{\text{fp}}$ in security. On a true attack the correct action is escalate; an explanation that reads benign induces suppress and incurs C_{miss}. Under groundedness failure the suppress event occurs with a seed-dependent probability $s(x)$, so the expected excess cost of trusting the explanation is $\mathbb{E}_x[s(x)]\, C_{\text{miss}}$ over the true attacks, and it concentrates, by Fig. 6, on the high-volume DoS and reconnaissance flows that dominate the attack set. Trusting the explanation therefore does not merely add noise; it adds expected cost weighted toward high-rate threats.

The blind spot is attack-type-dependent. Resolving the harm by attack type on the multiclass datasets (Fig. 6) shows it is far from uniform. On CICIDS2017 the divergence is largest on DoS (0.86) and port scanning (0.62), the high-volume denial and reconnaissance flows an analyst most needs to escalate, and smaller on DDoS (0.20). The pattern differs on TON-IoT, where divergence is high on some flooding and man-in-the-middle traffic but low on several stealthier types, so there is no guarantee that the seed-dependent blind spot spares the serious threats: on the most operationally realistic corpus it clusters precisely on the high-rate DoS and scanning attacks. Mis-founded trust does not spread its risk evenly, and it offers no assurance of sparing the attacks that matter.

VII. Can an Adversary Exploit This?

The failures above are benign: they arise without an adversary. Security demands the adversarial case. Fig. 8 contrasts two routes to get a malicious flow's alarm dismissed: the classical one evades the *detector*; the one this section studies keeps the detector firing but manipulates the *explanation* so the analyst dismisses a true alarm, an evasion by explanation rather than by detection. In vision and NLP the latter is established: explanations can be moved almost arbitrarily while the prediction is fixed [24], [25], [26].

TABLE XI
Explanation-induced harm on correctly-detected true attacks (logistic / random-forest auditor). Detection is stable; the analyst action induced by the explanation is not.

	UNSW	CICIDS	5G	TON	CIC23
detection agreement	0.961	0.880	0.998	0.995	0.986
action divergence	.12/.03	**.48**/**.87**	.02/.68	.37/.81	.00/.01
true attack dismissed	.04/.01	.29/.35	.00/.24	.10/.22	.00/.00

We tested the analogue on our tabular IDS. We model an analyst who reasons from explanations as an auditor classifier trained to map an attribution vector to benign/attack, and we search for a bounded perturbation that keeps the detector's malicious decision while steering the explanation toward the benign prototype. The attack was inconclusive: the detector kept firing on 1.0 of flows, but the manipulated explanations were still read as attack by the auditor almost everywhere, and the fraction of true alarms we could get dismissed never exceeded 0.26, most of which reflects the auditor's baseline error rather than the attack. A second formulation, directly driving attribution mass off the top-k features, also failed to move them.

The tabular result is not evidence that tabular IDS explanations are robust. The gap from the vision result is real and informative: tabular NIDS features are low-dimensional and semantically constrained, so the room to reshape an explanation without either flipping the decision or leaving the data manifold is small, unlike the high-dimensional redundancy of images. Stronger, model-agnostic attacks in the style of adversarial scaffolding [26] remain untested here and are a clear next step. We report the negative result because a security paper should not overstate an attack, and because the benign failures of Section V already void trust without needing an adversary.

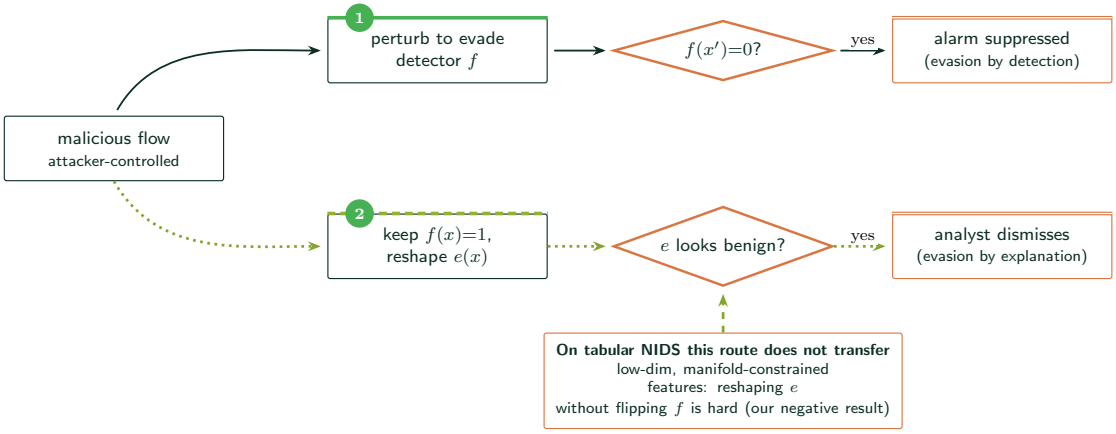


Fig. 8. Two routes to suppress a true alarm. Route 1 evades the detector (classical adversarial evasion). Route 2, ours, keeps the detector firing but reshapes the explanation so the analyst dismisses it. On tabular NIDS, Route 2 does not transfer from vision: low-dimensional, manifold-constrained features leave little room to move the explanation without flipping the decision.

VIII. Is Trust Cheaply Recoverable? A Negative Result

If interpretability-based trust fails by default, the natural hope is a cheap fix: a per-alarm *reliability gate* that tells the analyst *when* an explanation may be trusted, from the single deployed model, without the expensive machinery of an ensemble. We built the most natural such gate and report an honest negative: it looks good in-distribution but is circular and does not generalize, so trust is not cheaply recovered. We detail it because the way it fails is instructive about why the problem is hard.

What "reliable" means. We call an explanation reliable for a flow if it is *stable*: its top-k features are reproduced both across an ensemble $\{f_1, \ldots, f_E\}$ of equally-good models (groundedness holds locally) and across the attribution family $\mathcal{M}$ (determinacy holds locally). We combine the two into a per-flow reliability score, where $\bar{J}$ denotes the mean pairwise top-k agreement over the indicated set,

$$S(x) = \tfrac{1}{2}\,\bar{J}_{\text{model}}(x) \;+\; \tfrac{1}{2}\,\bar{J}_{\text{method}}(x), \tag{5}$$

with $\bar{J}_{\text{model}}$ taken over the ensemble $\{f_i\}$ (local groundedness, P3) and $\bar{J}_{\text{method}}$ over the family $\mathcal{M}$ (local determinacy, P2). $S(x)$ is the ground truth we want but cannot compute per alarm in deployment, since it needs an ensemble and several explainers. We stress that S measures *stability, not correctness*: a stable explanation is reproducible, not necessarily faithful, so the gate below recovers a necessary component of trust, not all of it (Section X).

The gate. We train a lightweight detector to predict this stability from signals that a single deployed model exposes at alarm time: the decision margin, the concentration (entropy) of the attribution, a local linear-surrogate fidelity, and an out-of-distribution distance (Algorithm 2). If the cheap detector predicts the expensive stability well, the analyst could be told, for free, when not to trust an explanation.

Algorithm 2: The reliability gate (which we show does not deploy)

Input: deployed model f; ensemble $\{f_i\}$; method family $\mathcal{M}$; training flows
Output: per-alarm reliability predictor h
`// offline: expensive stability label`
foreach *training flow* x **do**
  $S(x) \leftarrow \frac{1}{2}\bar{J}_{\text{model}}(x) + \frac{1}{2}\bar{J}_{\text{method}}(x)$ (Eq. 5);
`// cheap per-model signals`
$\phi(x) \leftarrow$
  [margin, attr. entropy, local fidelity, OOD dist.];
$h \leftarrow$ classifier fit on $\big(\phi(x), [S(x) > \text{median}]\big)$;
return h ; `// at alarm time:`
  `gate`$(x) = h(\phi(x))$

In-distribution, it looks promising. Fit and evaluated on the same dataset, the gate predicts explanation reliability with AUC 0.78 (UNSW-NB15), 0.85 (CICIDS2017), 0.95 (5G-NIDD), and 0.86 (TON-IoT) (Fig. 9), and out-of-distribution distance alone is weak (AUC 0.55 on CICIDS2017, near chance), so the signal is not simply an OOD detector. Taken alone this looks like a working, deployable fix. It is not, for two reasons.

Failure 1: it is circular. The signal that carries the gate is the attribution *concentration*, whose single-signal AUC (0.78 on 5G-NIDD, and 0.87 on TON-IoT, which *exceeds* that dataset's full-gate AUC of 0.86) matches or exceeds the full gate. But concentration is coupled to the label *by construction*: a peaked saliency vector yields a stable top-k set under both re-seeding and method change, so it raises both the reliability label $S(x)$ and the concentration feature (their correlation is ≈ 0.82 in magnitude). The gate is thus, in large part, the same quantity measured twice, not a cheap predictor of an independent property. The "not just OOD" signal is not independent of OOD: it is replaced by a feature entangled

with the target. Proposition 3 makes the entanglement precise.

Proposition 3 (Gate circularity)**.** *Let the concentration signal be the negative entropy of the normalized attribution,* $\kappa(x) = \sum_j p_j(x) \log p_j(x)$ *with* $p(x) = |e(x)|/\|e(x)\|_1$ *(so* $\kappa = -H$*, high when peaked). As* $\kappa(x)$ *increases, the attribution mass concentrates on fewer coordinates, so* $\mathrm{top}_k(e, x)$ *is increasingly determined by those coordinates and hence increasingly invariant to small changes of method or seed. Therefore* κ *is monotonically associated with both terms of* $S(x)$ (5)*, and a predictor using* κ *recovers* S *in part by definition, not by learning an independent cue.*

Empirically κ correlates with S at magnitude ≈ 0.82, consistent with the proposition.

Failure 2: it does not generalize. A deployable gate must work without being re-fit on the target dataset's expensive label. It cannot: trained on one dataset and tested on another, its AUC drops to a cross-dataset mean of 0.62, far below its in-distribution 0.78–0.95, and falls *below chance* on several pairs (e.g. 0.47 for UNSW→CICIDS, 0.47 for CICIDS→5G-NIDD, and 0.20 for CICIDS→TON-IoT); the fitted signal weights even flip sign across datasets. So the only way to make the gate work is to compute, per dataset, the very ensemble-and-multi-method reliability label it was meant to spare the operator. The cheap gate is not cheap.

Failure 3: it is adversarially gameable. A security control invites an adversary. Two of the gate's four inputs (decision margin, attribution entropy) are shaped by the input, so an attacker holding a genuinely-unreliable malicious flow can perturb it to make the gate accept its explanation as reliable while keeping the attack decision. This adaptive attack succeeds sharply: the fraction of low-stability true attacks the gate passes as reliable rises from 0.01 to 1.0 on CICIDS2017 (gain 0.99), with the malicious decision preserved throughout. An operator relying on the gate would be handed manipulated "reliable" explanations for real attacks.

Takeaway. The most natural reliability gate recovers a stability signal that is partly tautological, wholly non-transferable, and adversarially gameable. This indicates that recovering interpretability-based trust is a genuine open problem, not a matter of bolting on a confidence score. Whether a gate that predicts *faithfulness* (not mere stability) and generalizes across deployments can recover the warrant is untested and unsolved; we leave it to future work.

IX. Threat-Aware Evaluation Criteria

If interpretability is to warrant trust in a security setting, a claim to that effect should be held to checkable conditions. We propose the following, each the operational counterpart of a property or result above, and each currently unmet by typical explainable-IDS papers.

1) **Fidelity budget.** Report the decision agreement between the interpretable artifact the analyst sees and the deployed model, on held-out and on attack traffic. A trust claim below a stated threshold is unsupported.

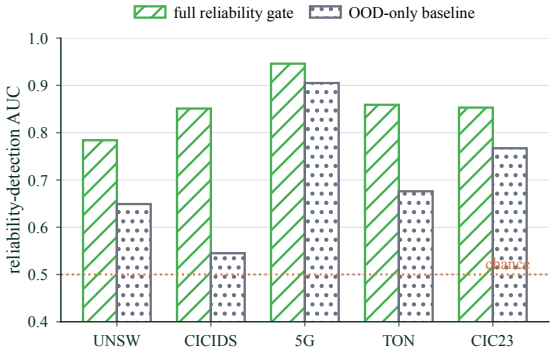


Fig. 9. In-distribution, the gate predicts explanation reliability with high AUC and beats an out-of-distribution baseline. This is misleading: the carrying signal is circular and the gate does not transfer across datasets (Failures 1–2). In-distribution success is necessary but far from sufficient for a deployable gate.

TABLE XII
IN-DISTRIBUTION RELIABILITY-GATE AUC: FULL DETECTOR VERSUS EACH SINGLE SIGNAL. OOD DISTANCE ALONE IS WEAK, BUT THE CONCENTRATION SIGNAL THAT REPLACES IT IS COUPLED TO THE LABEL BY CONSTRUCTION (FAILURE 1), AND NONE OF THIS TRANSFERS ACROSS DATASETS (FAILURE 2).

	UNSW	CICIDS	5G	TON	CIC23
full gate	**0.78**	**0.85**	**0.95**	0.86	0.85
margin only	0.60	0.58	0.70	0.65	0.75
concentration only	0.75	0.51	0.78	**0.87**	0.71
local fidelity only	0.62	0.61	0.69	0.24	0.75
OOD distance only	0.65	0.55	0.91	0.68	0.77

2) **Method invariance.** Report top-feature agreement across at least two standard attribution methods, including a perturbation-based one (SHAP or LIME). If the shown reason changes with the method, disclose it; do not present one method's output as *the* explanation.
3) **Instance invariance.** Report explanation agreement across performance-equivalent retrainings. Explanations that vary across equally-good models must not be sold as task insight.
4) **Adversarial robustness of the explanation.** State a threat model for the explanation itself and evaluate whether an adversary preserving the malicious decision can alter the shown reason. Absent this, interpretability is a usability feature, not a security one.
5) **Trust as warrant, not satisfaction.** Evaluate whether acting on the explanation agrees with ground truth, not whether analysts *feel* more confident.
6) **Reliability signalling.** Where per-flow reliability varies, provide a gate that tells the analyst when the explanation should not be trusted, and validate it.

These are necessary, not sufficient; but a claim that fails any of them should not be stated as a security benefit. Their cost varies: the method- and surrogate-agreement checks are inexpensive, whereas the model-instance (groundedness) check requires training a small ensemble, the same cost

that motivates, but does not rescue, the reliability gate of Section VIII. We release code that computes all six.

A. How current work measures up

To test whether these criteria are already met, we audit a corpus of twelve recent explainable-IDS papers (2020–2026) against the six (Table XIII, all twelve listed). *Selection.* We searched Google Scholar and arXiv for explainable/interpretable network-intrusion detection using SHAP, LIME, or attribution methods, published 2020–2026, and took the twelve most-cited or most-recent that build or evaluate an XAI-IDS; this is a convenience corpus, not an exhaustive survey, so we make no field-prevalence claim beyond it. *Coding.* We score each criterion from each paper's abstract and method only: a criterion is *reported* (1) only when the paper explicitly presents the corresponding evidence, *partial* (0.5) when it is touched but not measured (e.g. two explainers shown without an agreement number), and *not-reported* (0) otherwise; coverage is the mean score over the twelve (e.g. $C_1{=}(0.5{+}0.5)/12{=}8\%$). Scores are thus a lower bound on rigor, and the audit measures *reporting practice*. The full list with DOIs and per-criterion coding is released as `field_audit.csv`.

Table XIII shows the result, and it is stark. No criterion is met by a majority. Fidelity budgets are reported by 8% of papers, instance invariance by 0% (none), an explanation threat model by 12%, warrant-style trust evaluation (versus analyst satisfaction) by 12%, and a reliability signal by 4%. Only method invariance reaches 33%, and even there most papers that use two methods (SHAP and LIME) present them as complementary for *cross-verification* [55], not as a determinacy check, so agreement is assumed rather than measured. The applied papers, which build XAI-IDS systems and report analyst usefulness [56], [57], report essentially none of the six; the two that engage explanation quality directly are the evaluation and systematization works [17], [18]. In short, the properties whose failure we document are, in current practice, almost never checked, so the failures go unnoticed and the trust claims stand unchallenged.

TABLE XIII
AUDIT OF RECENT EXPLAINABLE-IDS WORK AGAINST THE SIX CRITERIA (● REPORTED, ◐ PARTIAL, ○ NOT REPORTED; CONSERVATIVE LOWER BOUND FROM ABSTRACT/METHOD). NO CRITERION IS MET BY A MAJORITY; INSTANCE INVARIANCE AND RELIABILITY SIGNALLING BY ALMOST NONE.

Paper	C1 fid	C2 m-inv	C3 i-inv	C4 adv	C5 warr	C6 rely
Warnecke [17] (eval)	◐	●	○	◐	◐	○
Nadeem [18] (SoK)	◐	◐	○	◐	◐	○
Capuano [1]	○	○	○	○	○	○
Le [23]	○	○	○	○	◐	◐
Uccello [57]	○	●	○	○	○	○
Muhammad [55]	○	○	○	○	○	○
Farrukh [56]	○	○	○	○	○	○
Barnard [58]	○	◐	○	◐	○	○
Sharma [59]	○	◐	○	○	○	○
Mohale [60]	○	◐	○	○	○	○
Dong [61]	○	○	○	○	○	○
Faqihi [62]	○	○	○	○	○	○
coverage	8%	33%	0%	12%	12%	4%

X. DISCUSSION

Accountability. Operational security increasingly places explanations in the decision loop and, implicitly, in the accountability chain: an analyst who acts "because the system showed feature X" is relying on e. Our results imply this reliance is, today, reliance on an artifact of tool and seed. Accountability regimes that credit "the model is explainable" as a control are mis-specified unless the explanation meets conditions like ours; and, as our negative result shows, one cannot yet discharge that condition with a cheap per-alarm reliability score.

Why recovering trust is hard. A tempting fix is to score, per flow, how much to trust an explanation. Our negative result (Section VIII) shows why the natural version does not work: stability is measurable but partly tautological with its own predictors and does not transfer across deployments, and stability is not faithfulness in any case. A useful gate would have to predict *faithfulness* and generalize, which post-hoc attributions on expressive models may not permit. Selective trust is the right idea; a sound mechanism for it is unsolved.

Scope. The *inference* from interpretability to trust is unlicensed until the properties are checked, and on modern data the check fails by default; this is not specific to the particular methods we test. We report the adversarial manipulation result as inconclusive and the reliability gate as a negative result.

Threats to validity. *Attribution scope*: we study five gradient- and perturbation-family attributions and tree/logistic/KAN surrogates; prototype-, concept-, and rule-based methods [9] may behave differently, which is exactly what our criteria ask authors to check, but the fidelity, faithfulness, and cross-family determinacy failures are method-diverse enough that a single better method is unlikely to rescue trust. *Analyst model*: the analyst is a classifier, not a human; a real analyst uses context beyond the attribution vector and would be swayed less by it, so our simulated divergence is an *upper bound* on a real analyst's explanation-induced divergence, and a human study is essential future work before any operational claim. *Detector scope*: beyond the MLP and KAN detectors, we re-ran determinacy, groundedness, and faithfulness on the tree ensembles that dominate tabular NIDS, random forests and gradient boosting explained by *exact, deterministic* TreeSHAP (Table VII); all three still fail in the same range, so the result is not an artifact of neural detectors or of Monte-Carlo explainers. A convolutional or transformer NIDS remains future work. *Dataset difficulty*: 5G-NIDD is near-saturated in detection, which is why its harm is small; the effect is concentrated, as it should be, on the harder CICIDS2017 and TON-IoT. *Statistical power*: we report three-seed statistics and, for the gate, expose its true run-to-run instability rather

than an over-tight interval. None of these points toward a hidden way that interpretability *does* warrant trust; they bound where our specific numbers apply.

XI. Conclusion

Explainable ML-IDS invites analysts to trust detectors through their explanations. We asked what would make that trust well-founded, formalized it as fidelity, determinacy, and groundedness, and showed, on five modern intrusion datasets, that fidelity fails for the interpretable-surrogate route and that determinacy and groundedness fail across five standard explainers including SHAP and LIME. We showed the failure can carry a security cost: on correctly-detected true attacks, and at a dataset-dependent rate that reaches 0.87 of flows on CICIDS2017 (near-zero on the easiest task), equally-good detectors' explanations drive opposite analyst actions and there make a true attack read benign 0.29 to 0.35 of the time (a simulated-analyst upper bound), so where the effect is strong, acting on the explanation makes catching a real attack an accident of the deployed model. We then mapped how hard the failure is to fix, reporting two honest negatives, an explanation-manipulation attack that does not transfer to tabular IDS, and a natural reliability gate that is circular, non-transferable, and adversarially gameable. Interpretability, as currently produced and evaluated, is not a sound basis for trust in ML-IDS, and recovering that basis is an open problem. We offer threat-aware criteria so that the next claim of interpretability-for-trust in security is made, and checked, on evidence.

Appendix A
Full Pairwise Explanation Agreement

Table XIV gives the complete top-5 Jaccard agreement for all ten pairs of the five explainers, at the high sampling budgets used throughout (KernelSHAP 1000 samples, LIME 3000). The two variants within the gradient family that share the input×gradient construction (input×gradient vs. integrated gradients) are the only consistently high pair (0.63–0.83); every cross-family pair is low. This is why we anchor the determinacy claim on cross-family disagreement (e.g. gradient vs. SHAP, 0.17–0.33) rather than on the noisier SHAP-vs-LIME comparison: at low sampling budgets SHAP and LIME are not even self-consistent (rerun agreement ≈ 0.43), so part of their mutual disagreement is Monte-Carlo noise, whereas the cross-family gap persists at any budget.

TABLE XIV
Top-5 Jaccard agreement for all explainer pairs (high sampling budget). G=gradient, I=input×grad, IG=integrated grad, S=SHAP, L=LIME.

Pair	UNSW	CICIDS	5G	TON	CIC23
g/i	0.36	0.22	0.51	0.30	0.39
g/ig	0.34	0.20	0.51	0.26	0.42
g/s	0.24	0.17	0.31	0.26	0.33
g/l	0.20	0.24	0.50	0.21	0.37
i/ig	0.75	0.78	0.82	0.63	0.83
i/s	0.41	0.55	0.50	0.49	0.21
i/l	0.19	0.20	0.49	0.15	0.22
ig/s	0.40	0.58	0.51	0.51	0.24
ig/l	0.18	0.20	0.50	0.21	0.25
s/l	0.37	0.25	0.36	0.18	0.31

Appendix B
Reproducibility

We release the complete artifact, the five datasets' preprocessing, every experiment script, and the figure and table generators, so each quantitative claim in the paper can be regenerated from raw data. The anonymized artifact is at https://anonymous.4open.science/r/xai-ids-trust-2826; a permanent, DOI-archived mirror will replace it in the camera-ready. Table XV maps every result to the script that produces it and the file it writes, and each number in the paper is injected at compile time from a one-value-per-file store (`paper/claims/`), so no figure in the text is transcribed by hand.

Environment. Python 3.9.6 with `numpy` 2.0.2, `scipy` 1.13.1, `scikit-learn` 1.6.1, `torch` 2.8.0, `shap` 0.49.1, and `lime` 0.2.0.1, all pinned in `requirements.txt`. Every experiment fixes the `numpy` and `torch` seeds before loading data, so a re-run reproduces the reported numbers exactly; we verified determinism by double-run comparison.

Data. All five datasets are public: UNSW-NB15 [49], the cleaned CICIDS2017 [50], [51], 5G-NIDD [52], TON-IoT [53], and CIC-IoT-2023 [54] (Table III). `data.py` applies the identical pipeline to each: binary normal-versus-attack labels, categorical encoding, standardization, a 40k stratified subsample, and a 70/30 stratified split, dropping identifier columns (IP addresses, ports, timestamps) so an attribution is a claim about behaviour, not host identity.

Running it. After `pip install -r requirements.txt`, running `python <script>.py` for any row of Table XV regenerates that file, and `python plots.py` rebuilds every figure. The full suite completes in well under an hour on a commodity CPU; no GPU is required.

Appendix C
Ethics Considerations

This work studies public network-intrusion datasets (UNSW-NB15, CICIDS2017, 5G-NIDD, TON-IoT, CIC-IoT-2023) and introduces no new data collection or human-subjects experiment; the analyst in our study is modeled by a classifier, not a person, so no IRB approval was required. The paper is a defensive contribution: it warns that a widely promoted security workflow, trusting ML-IDS through explanations, may be unfounded, and offers a mechanism to mitigate the risk. Our inconclusive attack is described at a level sufficient to reproduce the analysis but does not provide an operational tool for evading detectors, and the underlying benign failures are the primary basis of our conclusions. We release code, the data-preparation scripts, and the evaluation criteria to help the community audit interpretability-for-trust claims, which we judge net-beneficial: the same code that could reveal a

TABLE XV
EVERY RESULT MAPS TO THE SCRIPT THAT PRODUCES IT AND THE OUTPUT FILE IT WRITES. NUMBERS IN THE PAPER ARE INJECTED FROM PAPER/CLAIMS/, ONE VALUE PER FILE. SCRIPT AND FILE NAMES OMIT THE .PY AND .CSV SUFFIXES.

Result	Script(s)	Output
P1–P3 (Tab. IV, Fig. 2)	exp_trust, exp_multiseed	trust, multiseed
Fidelity frontier (Tab. V, Fig. 3)	exp_fidelity_multi	fidelity_multi
Determinacy, 5 explainers (Fig. 4, Tab. XIV)	exp_breadth	breadth
Redundancy robustness (Tab. VIII)	exp_redundancy	redundancy
k/ensemble robustness (Fig. 5)	exp_ablate_p2p3	ablate_p2k, ablate_p3E
Positive control (Tab. VI)	exp_ceiling	ceiling
Tree ensembles (Tab. VII)	exp_tree	tree
Faithfulness (Tab. X, IX)	exp_faithfulness, exp_faith_5explainer	faithfulness, faith_5explainer
Security harm (Tab. XI, Fig. 7, 6)	exp_harm, exp_harm_bytype	harm, harm_bytype
Reliability gate, ablation, transfer (Tab. XII, Fig. 9)	exp_defense, _ablate, _transfer, _gate_attack	defense, defense_ablate, defense_transfer, gate_attack
Manipulation attack (non-transfer)	exp_attack	attack
Field audit (Tab. XIII)	field_audit	field_audit

weak explanation is what a defender needs to avoid over-trusting one. The artifact is available at https://anonymous.4open.science/r/xai-ids-trust-2826 (anonymized for review; a non-anonymous mirror will replace it in the camera-ready). We disclose no specific vulnerable product; the concern is a methodological one affecting the class of explainable ML-IDS.

Use of generative AI. Per the submission policy we disclose that generative AI assistants were used while preparing this work, in two roles: editing and condensing prose, and writing auxiliary tooling (the reference-verification and number-tracing scripts that check the manuscript against the released CSVs). They were not used to generate experimental results, and every number reported here is produced by the released code and mechanically checked back to a result file. The authors take full responsibility for all claims, results and citations.

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